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Irrigation effects on the growth, yield, and water use efficiency of alfalfa

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Abstract A field study was conducted in the semiarid region of northern Sudan to investigate the effects of variable irrigation on the growth, yield, and water use efficiency (WUE) of alfalfa (*Medicago sativa* L.). Treatments were 65 mm of water applied every 7 days, 80 mm of water applied every 10 days, or 104 mm of water applied every 13 days. The heavy, infrequent irrigation reduced stem height, stem density, leaf area index (LAI), total biomass production, and the WUE of alfalfa plants. Maximum yields for six harvests were 15.3, 12.9, and 11.2 ton ha⁻¹ and the WUE values were 0.12, 0.10, and 0.08 ton ha⁻¹ cm⁻¹ for the frequent, less-frequent, and infrequent irrigation regimes, respectively. In all the treatments, alfalfa dry matter yield was positively correlated with stem height and LAI. The relationship between dry matter yield and total water use was a linear function ($R^2=0.99$), regardless of the irrigation treatment. Alfalfa growth, yield, and WUE remained high during the relatively cool months and declined during the hot period under the three water regimes. It was concluded that alfalfa grown under semiarid conditions should be watered lightly and frequently to attain high yields and high WUE.

Introduction

Data on the water use of crops grown in semiarid climates are essential for optimum irrigation management strategies and water conservation. Alfalfa (*Medicago sativa* L.) has a high water requirement compared to other crops (Krogman and Hobbs 1965; Blad and Rosenberg 1976). Several studies have shown that soil water deficits resulted in re-

duced alfalfa growth and yields (Kisselbach et al. 1929; Lucey and Tesar 1965; Brown and Tanner 1983; Grimes et al. 1992). Alfalfa stem density (Cowett and Sprague 1962; Cohen et al. 1972; Bogler and Matches 1990), stem height (Jensen et al. 1967; Brown and Tanner 1983; Bogler and Matches 1990), and leaf size (Gindel 1968) decreased when soil water deficits developed.

A number of reports have shown a linear relationship between dry matter (DM) production and water use for alfalfa (Bauder et al. 1987; Sheaffer et al. 1988; Grimes et al. 1992). However, this relationship may vary from one climatic region to another (Hill et al. 1983), from one season to another, and from one harvest to another during the growing season (Myer et al. 1991). The slope of the linear relationship between biomass production and depth of water used is frequently expressed as the water use efficiency (WUE). Alfalfa WUE reported in the literature ranges between 0.11 and 0.23 ton ha⁻¹ cm⁻¹ with the greater values in the cooler, more northerly climates (Kisselbach et al. 1929; Stanhill 1986; Bogler and Matches 1990; Grimes et al. 1992).

Data on the effects of variable irrigation on alfalfa growth, yield, and water use under semiarid conditions are insufficient. In northern Sudan, alfalfa production is heavily dependent on the availability of water supplied mainly through irrigation from the Nile River. Irrigation strategies that result in higher yields and efficient use of water are needed in order to reduce the cost of alfalfa production.

Our objectives were to study the growth and yield responses of alfalfa to three contrasting irrigation treatments and to determine the water-yield relationships and WUE under these regimes.

Materials and methods

This study was conducted on the Demonstration Farm of the Faculty of Agriculture, University of Khartoum (15°40' N, 32°32' E), Sudan. The climate of the area is semiarid and tropical (Oliver 1965). The main features of the climate are the relatively cool winter, hot summer, highly irregular rainfall, and a potential evapotranspiration

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Table 1 Monthly averages of some meteorological data at Shambat, Sudan during the period May 1982–July 1983 (*RH* relative humidity, *TR* trace)

Month	Air temp. (°C)		RH	Rainfall	Pan evaporation
	min.	max.			
			(%)	(mm)	(mm)
May	24.4	41.1	21	9.7	18.7
Jun	26.2	41.3	29	TR	13.7
Jul	25.8	39.9	43	6.8	11.4
Aug	25.5	37.3	61	23.3	8.8
Sep	25.9	39.3	52	34.3	9.3
Oct	24.3	38.2	37	TR	10.5
Nov	16.7	32.5	30	0.0	10.2
Dec	13.9	30.4	35	0.0	8.7
Jan	11.0	24.9	34	TR	8.4
Feb	14.1	31.0	27	0.0	9.7
Mar	16.0	33.2	22	TR	11.0
Apr	20.3	38.1	17	0.0	13.0
May	24.4	42.1	21	0.0	12.8
Jun	27.7	40.8	41	12.3	12.7
Jul	26.1	40.1	50	11.4	10.8

that exceeds precipitation throughout the year. Average annual rainfall is about 160 mm with great variation in the interval between showers and amount of rain per shower, as well as the length of the rainy period. Temperature, due to its high variation, is the main climatic factor affecting crop water requirement in the different seasons (Saeed 1984). Table 1 shows some of the meteorological data from Shambat Meteorological Station located about 400 m from the experimental plots.

The farm was on the low terrace of the river Nile. The soil is of alluvial origin deposited by the river and is characterized by a high clay content (40%), poor structure, low organic matter content, low permeability, and high water-holding capacity (El-Attar 1969). The gravimetric soil water contents at -0.3 (field capacity) and -15 bar (permanent wilting percentage) were 54.3 and 16.1%, respectively (Dover 1966).

Alfalfa (cv. Berseem Higazi) was planted on the flat in 16-m^2 plots at a seeding rate of 40 kg ha^{-1} . A basal dressing of nitrogen fertilizer in the form of urea at a rate of 95 kg ha^{-1} was added at the time of planting. Presowing irrigation was applied to the experimental plots in order to achieve uniform water distribution in subsequent irrigations and to control weeds before planting. The layout of the experiment was a completely randomized block design with four replications.

The borders of the experimental plots were excavated to a depth of 90 cm where overlapping galvanized iron sheets were installed and buried to the ground level to restrict lateral water movement (El-Nadi 1969). Earthen embankments 60 cm high separated the plots at the soil surface. The irrigation water was introduced by an irrigation setup designed by El-Nadi (1969), which consists of a water reservoir, an electrical pump, and a single main pipeline with branches to serve the experimental plots. At the point of delivery, a gate valve and a pressurized rubber attachment were installed to allow the delivery of water to each plot. Variable irrigation was initiated after the first harvest of the crop (75 days after planting). Three irrigation regimes were used: treatment A (65 mm applied every 7 days), treatment B (80 mm applied every 10 days), and treatment C (104 mm applied every 13 days). These treatments will be referred to as light frequent, moderate less-frequent, and heavy infrequent regimes, respectively. The amount of water applied in each treatment was equivalent to 8 mm per day. This value was the maximum crop evapotranspiration reported for both cotton and short grass in the same region (Fadl 1978, 1979). The total number of irrigations during a period of 6 months was 25, 18, and 13 for treatments A, B, and C, respectively. The total depth of water applied per treatment during the 6-month period was 1400 mm. However, in order to apply equal total water inputs in each treatment, 40 mm of water were sub-

Table 2 Dry bulk density of the different soil depths in the demonstration farm at Shambat, Sudan

Soil depth (cm)	Bulk density (g cm^{-3})
0 – 15	1.25
15 – 30	1.37
30 – 45	1.38
45 – 60	1.42
60 – 75	1.51
75 – 90	1.58

tracted from the last irrigation in treatment B, and 48 mm of water were added to the last irrigation in treatment C.

Crop evapotranspiration was estimated using the water balance equation. Where surface runoff or runoff has been eliminated by earthen embankments, the equation can be written as

$$Et = I - D \pm S_w$$

where Et is the crop evapotranspiration (mm), I is the water input by irrigation and/or rainfall (mm), D is deep drainage (mm), and S_w is the soil water depletion (mm).

Deep drainage was not measured directly. It was evaluated graphically (Grimes et al. 1992) by plots of cumulative DM yield as a function of total water from $I \pm S_w$. This relationship was linear until $I \pm S_w$ exceeded about 1200 mm. Apparent Et ($I \pm S_w$) > 1200 was not associated with yield and is, therefore, assumed to result from D . Drainage was taken into account by excluding data from DM yield- Et analysis, where $I \pm S_w$ exceeded 1200 mm. Soil water depletion (S_w) was calculated from soil water profiles measured from 0–15, 15–30, 30–45, 45–60, 60–75, and 75–90 cm depth ranges prior to each irrigation by the gravimetric method. The following expression was used to calculate the gain or loss in soil water (S_w) in the profile

$$S_w = (M_2 - M_1) \text{pd} \cdot d / 100$$

where M_1 and M_2 are the initial and final gravimetric soil water contents (%) in the soil layer, respectively, pd is the soil bulk density (g cm^{-3}), and d is the thickness of the soil layer (cm). Soil bulk densities were determined for the soil by the clod method (Shaw 1917) and are given in Table 2.

Plot harvesting was scheduled every 4 weeks following a harvest and was continued for 6 months. Plant dry weights were determined by harvesting and drying (70°C for 48 h) plant samples cut from the 16-m^2 plots. Subsamples from an area of 0.09 m^2 from the center of each plot where harvested every 10 days to determine stem density, stem growth rate, final stem heights, and leaf area. Leaf area was estimated using the punch method (Watson and Watson 1953) and the leaf area index (LAI) was calculated accordingly.

WUE was calculated at each plant sampling date by dividing plant DM weights (ton ha^{-1}) by total water used (cm). Total water use during a growth period was the sum of water extracted by the crop plus rain and/or irrigation received during that period minus the estimated deep drainage.

Statistical analysis was performed on growth attributes, yield, and soil water data using SAS (SAS 1987). Mean separation was performed using least-square differences (LSD).

Results

The results presented are from six 30-day regrowth periods from 1 February through 31 July 1983, named harvests one through six. Light, frequent irrigation resulted in a significantly ($P=0.05$) higher water content in the top soil than in the other treatments (Fig. 1). Treatment B resulted in intermediate levels of water content in the profile. Plots under treatment C had the lowest soil water contents

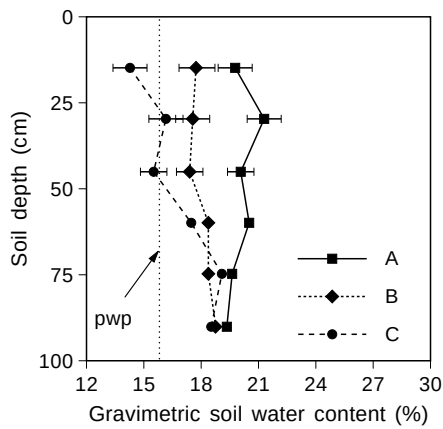


Fig. 1 Gravimetric soil water contents (%) at different depths under the three irrigation regimes averaged for the six harvests. Measurements were taken prior to each irrigation. Error bars are the least-square difference (LSD) at $P=0.05$. The permanent wilting percentage (pwp) was 16.1 and the field capacity was 54.3%

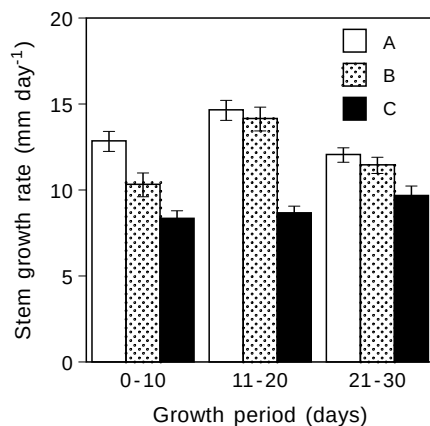


Fig. 2 Alfalfa stem growth rate (mm day^{-1}) for the three water regimes during the initial (0–10 days), intermediate (11–20 days), and final (21–30 days) regrowth cycles. Error bars are the LSD at $P=0.05$

(Fig. 1), which were at or below the permanent wilting percentage, especially in the top soil layers. It is also apparent from Fig. 1 that most of the soil water depletion by alfalfa occurred in the top 70 cm of the profile, where most of the difference in soil water content between treatments can be accounted for.

The stem growth rate was slow initially (0–10 days), increased during the intermediate growth period (11–20 days), and then decreased during the final 10 days of regrowth in treatments A and B. The severely stressed plants in treatment C had a constantly slower stem growth during the whole regrowth period compared to the other two treatments (Fig. 2).

In the final harvests of the six cuttings, treatment A resulted in taller plants, followed by intermediate and short plants for treatments B and C, respectively (Fig. 3). There were significant differences ($P=0.05$) in stem heights

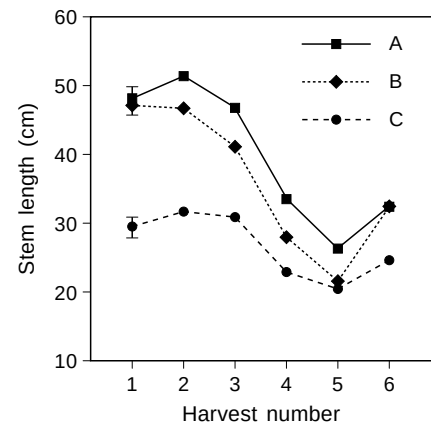


Fig. 3 Alfalfa stem height (cm) at the final harvest for each of the six cuttings of the various water regimes. Error bars are the LSD at $P=0.05$

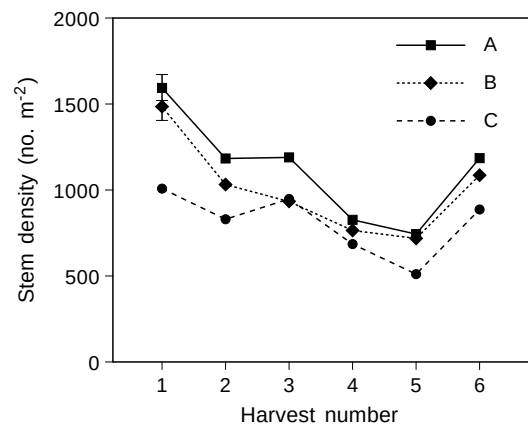


Fig. 4 Alfalfa stem density (number m^{-2}) for the six harvests under the three water regimes. Error bars are the LSD at $P=0.05$

between the frequent and infrequent irrigation regimes throughout the growing period. Plants under treatment B were significantly shorter than those under treatment A in four harvests, and were significantly taller from those under treatment C in five harvests (Fig. 3). Maximum and minimum stem heights under each regime were obtained in harvests two and five, respectively. The different irrigation treatments resulted in significant differences ($P=0.05$) in alfalfa stem density (number m^{-2} ; Fig. 4). Generally, a water deficit decreased stem density to 10–30% of that of frequently watered plants in the 6 months of regrowth (Fig. 4).

Figure 5 shows the LAI values for alfalfa plants in the initial, intermediate and final harvests for the three water regimes throughout the six regrowth cycles. The severely stressed plants in treatment C had the lowest LAI in three harvests. The intermediate irrigation treatment had little or

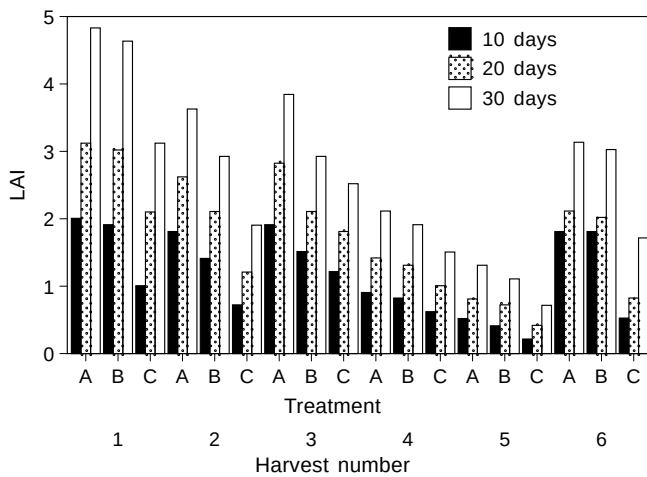


Fig. 5 Alfalfa leaf area index (LAI) measured at initial (0–10 days), intermediate (11–20 days), and final growth periods (21–30 days) for the six harvests under the three water regimes

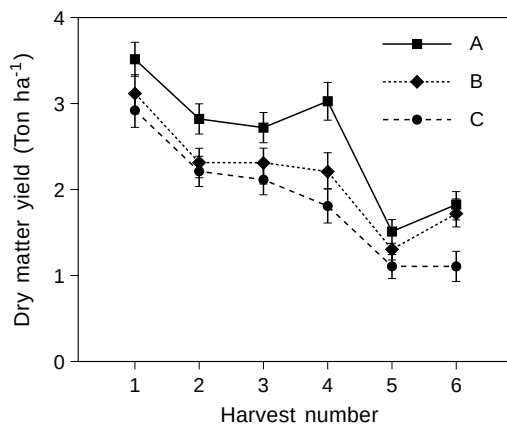


Fig. 6 Alfalfa dry matter yield (ton ha⁻¹) for the six cutting cycles under the three water regimes. Each values is an average of four measurements. Error bars are the LSD at $P=0.05$

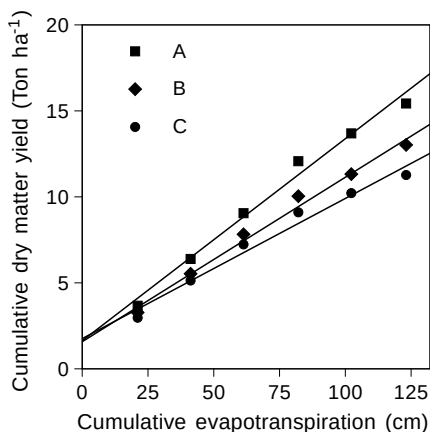


Fig. 7 Cumulative dry matter yield (ton ha⁻¹) vs. cumulative evapotranspiration (cm) of alfalfa for the three water regimes. Each value is an average of four measurements

Table 3 Regression coefficients for alfalfa total dry matter yield vs. cumulative evapotranspiration for the three water regimes. General equation is $\text{yield} = a + b \cdot \text{Et}$

Treatments	Regression coefficients		R^2	n
	a (SE)	b (SE)		
A	1.54 (0.05)	0.12 (0.01)	0.99	6
B	1.51 (0.07)	0.10 (0.01)	0.99	6
C	1.81 (0.09)	0.085 (0.01)	0.98	6

no effect on LAI. Frequently irrigated plants showed significantly ($P=0.05$) higher LAI throughout the growing period in comparison to water-stressed plants (Fig. 5). Plants under treatment B showed LAI values intermediate to the other two regimes.

DM production was consistently lower for the stressed plants of treatment C than for the frequently watered plants (Fig. 6). There was no significant difference ($P=0.05$) in DM production between treatment B and C in five harvests (Fig. 6). Alfalfa DM production was correlated with stem height ($R^2=0.77$) and LAI ($R^2=0.79$) regardless of the water regime. Regardless of the irrigation treatment, alfalfa DM production was negatively correlated with average monthly temperature ($R^2=0.73$). This was reflected in consistently higher productivity in the relatively cool than in the hot months.

The total DM yield versus cumulative water use for each treatment is shown in Fig. 7. The DM yield for each treatment increased essentially linearly with cumulative water use. The slopes of the lines give the DM WUE (ton ha⁻¹ cm⁻¹). These slopes and the correlation coefficients are presented in Table 3. The average WUE of alfalfa plants in treatment A was higher than that in treatment B and C by 16 and 33%, respectively.

Discussion

Our results demonstrated that stress was evident in the heavily and infrequently watered plots of treatment C as established by the lower availability of water in the profile compared to the other two treatments. Investigations by other researchers have shown that drought stress resulted in reduced plant growth (Brown and Tanner 1983). Drought inhibited cell elongation, reduced photosynthesis, interfered with nutrient uptake, and altered plant hormone levels (Antolin and Sanchez-Diaz 1993).

It was also evident from this study that heavy, infrequent irrigation reduced alfalfa stem heights throughout the study period. This finding was consistent with that of Hill et al. (1983), who reported reduced stem length of previously stressed compared with unstressed alfalfa during the first 14 days of growth. Stem growth was slow during the initial growth period, increased through the intermediate growth period and then decreased during the final growth period under treatments A and B. Stem growth was

significantly different between all the treatments initially, but was not different between treatments A and B during the intermediate and final growth periods. However, reduced stem growth early in the growth cycle was evidenced by shorter stems after 30 days of growth for treatments B and C (Fig. 3). These results agree with those of Lucey and Tesar (1965), Hill et al. (1983), and Brown and Tanner (1983), in that near-constant stem extension rates occur throughout the vegetative growth period.

Drought stress decreased the stem density of alfalfa plants during the six cycles of regrowth. These results agree with those of Tanner and Brown (1983) and Carter and Sheaffer (1983), but were not in conformity with the finding of Hill et al. (1983). The differences between our results and those of the latter researchers may be due to the fact that alfalfa plants in our experiment were subjected to a more severe water stress.

Significant differences in LAI were found between stressed and non-stressed plants throughout the sampling period. Water stress affected the leaf initiation process, number of leaves produced, and leaf blade size primarily through an effect on cell division (Berlin et al. 1981). Development of stress (2 weeks after cutting) reduced alfalfa leaf size but not leaf or internode number (Brown and Tanner 1983). Although several reports have shown an increased leaf/stem ratio due to water stress (Carter and Sheaffer 1983; Halim et al. 1989), our results showed a reduced LAI of stressed plants due to a reduction in crop density.

Results from this experiment indicated that a soil water deficit reduced DM yield, which decreased linearly ($R^2=0.90$) with increasing soil water deficit. Even the mild stress of treatment B caused a 15% reduction in DM yield. These results agree with those of Brown and Tanner (1983) and Hall (1993). The poor growth noticed during the months of May and June may largely be due to the high temperature during these months. The sensitivity of plants to increasing soil water stress increased under conditions of high temperature, low relative humidity, high wind velocity, and increased light intensities – all conditions which increase transpiration considerably.

Results from the three irrigation regimes were uniformly consistent in demonstrating a high degree of correlation ($R^2=0.99$) between DM yield and crop water use. These results concur with several earlier reports (Sammis 1981; Bauder et al. 1987; Grimes et al. 1992). The different slopes of these regression lines indicate the different WUEs under the three regimes. Differences in WUE in this experiment were attributable mostly to greater growth of the frequently watered plants. A water deficit, however, produced less WUE by reducing alfalfa growth and stand. When the DM-water use functions were extrapolated to zero yield they indicate that a maximum of 17 cm of water was used for DM production that was not accounted for in the cumulative water use as measured. This was contributed by the soil water stored at the time the treatments were initiated.

The WUE of the frequently watered alfalfa plants in this study was low compared to what has been reported in the

literature for cooler climates (Grimes et al. 1992), but was similar to that obtained by Sammis (1981) in warmer climate conditions. This could be attributable to the fact that yields of alfalfa grown under semiarid conditions were depressed due to higher temperatures. High temperature can increase water use of the crop and thus decrease its WUE (Saeed 1984).

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